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An Explainable Machine Learning Framework for Early Esophageal Cancer Prediction Using SHAP and LIME

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Abstract*—Esophageal cancer (EC) is a very aggressive cancerous disease, so detecting it early is crucial in enhancing its treatment and lowering the deaths. As opposed to traditional statistical approaches that have been adopted in EC risk assessment over the years, machine learning (ML) has not been explored extensively, even though it has better predictive properties. In this study, used a dataset of 3,985 patient records with 85 clinical and demographic variables to create an ML-based model. Seven classifiers were evaluated after extensive preprocessing and selection of features based on Gini Index. The K-Nearest Neighbors (KNN) model portrayed the highest accuracy of 96.84%. SHAP and LIME were used to ensure transparency and clinical reliability by ensuring global and local interpretability, which validates the impact of these characteristics on model predictions. The findings indicate that KNN model can be effectively used to support early risk stratification and focused screening. On the whole, the study identifies the promise of explainable ML methods in the increased detection of early EC and better, data-driven clinical decision-making.

Keywords*—Esophageal cancer, machine learning, feature importance, Gini index, KNN, Explainable AI.

I. INTRODUCTION

Esophageal cancer follows high prevalence levels and demonstrates aggressive behavior worldwide. Each year, almost 600,000 new cases are identified as EC globally [1]. According to reports, there are 15,850 EC-related deaths and 17,290 new cases in the US each year [2]. Although EC is frequently asymptomatic, at advanced stages it can present with a variety of symptoms, including dysphagia, weight loss, and odynophagia. Additionally, EC patients who receive an early diagnosis have a higher survival rate [3]. The increased frequency of EC and its dismal prognosis specify the need for better diagnostic and predictive strategies via adequate screening procedures. Achieving EC's preventative objectives requires it. Due to the aggressive and quiet nature of EC, which leads to a poor prognosis, early identification is essential to increasing the rate of survival after five years and reducing the mortality rate [4].

Endoscopy and biopsy methods can identify EC premalignant tumors such as squamous dysplasia. This solution is unfeasible since screening procedures based on these techniques are problematic for some high-risk categories of patients, such as elderly adults or those with gastrointestinal disorders [5]. Furthermore, the screening techniques used in communities, such as endoscopy, are expensive, intrusive, and have dubious efficacy in identifying high-risk EC individuals, which restricts their use at the population level [6]. To implement preventative measures and improve the quality of life for high-risk populations, prediction models are increasingly being used to evaluate disease risks for screening purposes [7].

Several studies have developed risk prediction models using statistical regression analysis to identify high-risk groups for EC as part of preventive strategies. Chen et al. built a logistic regression prediction model that detects individuals at high risk for developing EC. The predictive model included age, gender, and smoking status as well as symptoms (back pain) and dietary choices and history of upper gastrointestinal cancer in patient relatives. Risk prediction of EC showed an AUROC score value of 0.81 through the developed model [8]. Wang et al. developed a risk prediction model for EC using competing risk regression models. The researchers incorporated smoking and alcohol use alongside BMI measurements and physical activity data and demographic traits into their model. The model demonstrated AUROC performance of 0.76 for internal tests and 0.7 for external validation [9]. Etemadi et al. built a model that derived risk metrics from Lin regional factors, which involved water source and tea temperature alongside dental health and opium usage, together with demographic information. Their multivariate logistic regression model differentiated high-risk EC clusters while achieving an AUROC value of 0.77 [10].

Machine learning as a subfield of artificial intelligence (AI) received increased attention through growing digital data usage like Electronic Health Records (EHR) across various medical domains [11]. Nevertheless, stratifying the high-risk EC group requires the development of an effective risk prediction model [12]. Until now, the majority of previous studies have concentrated on creating EC prediction models using statistical techniques and risk factors, with minimal focus on machine learning in this regard. The key contributions of this paper:

- **Effective Feature Engineering Using Gini Index:** Comprehensive preprocessing and Gini Index based feature selection allowed the identification of the most influential clinical predictors, improving model efficiency by reducing noise in high-dimensional patient data.

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- **High Performance Prediction Model**: A robust machine learning framework was developed for early esophageal cancer detection, where the KNN classifier achieved the best accuracy of 96.84%, surpassing the performance reported in previous studies on EC predictions
- **Explainable AI for Clinical Transparency**: Integration of SHAP and LIME allowed for interpretability at both the global and instance level, increasing model transparency, clinical trust, and practical usability.

The remainder of this paper is organized as follows: Section II presents **Literature Review** of related work on esophageal cancer prediction; section III presents **Materials and Methods** which describes the dataset, preprocessing steps, feature selection, and machine learning methodology; section IV presents **Result Analysis** reports of the experiments, model comparison, and interpretability analysis with SHAP and LIME; Lastly, section V presents **Conclusion** to encapsulate the principal findings and identifies future research directions.

II. Related Works

Several studies have investigated the use of machine learning methods in predicting esophageal cancer outcomes. Nopour et al. created a risk model with six ML methods Random Forest, XGBoost, KNN, SVM, and ANN on a retrospective cohort of 3,256 cases in Iranian hospitals. XGBoost performed the highest with an AUROC of 0.92 on internal data and 0.83 on external data. The key predictive factors were the consumption of hot beverages, fruits, achalasia, and diet. The generalizability of the model was, however, undermined by it being regional in nature and employing mode-based imputation [13]. Rahman et al. constructed a model for predicting recurrence in EC surgical patients by combining clinical and histological features with an ensemble of Random Forest, XGBoost, and Elastic Net Regression (ELR). The best AUROC value of 0.805 belonged to the ensemble with strong external validation (AUROC = 0.804) [14]. Yoon et al. employed ML to predict the excessive loss of muscle in 232 male NACRT patients. The important features were platelet-to-lymphocyte ratio, albumin, neutrophil-to-lymphocyte ratio, nutritional statuses, and changes in BMI. Logistic regression and SVM combined were 0.808 in AUROC [15]. Gong et al. created a five-year survival prediction model using 10,588 EC cases from the SEER database. Different ML methods were attempted with GBM, CatBoost, LightGBM, Gradient Boosting Decision Trees (GBDT), Random Forest, Naïve Bayes, SVM, and ANN. XGBoost gave the highest performance with an AUROC of 0.852 through five-fold cross-validation [16]. Lin et al. proposed a semantic segmentation system of EC through the blend of ResNet and U-Net architectures for white-light and narrow-band endoscopic images. The system reached 85% and 82% accuracy on the narrow-band and white-light images, respectively. The system did not incorporate clinical or demographic data, and hence its use was limited [17]. Zhang et al. utilized EHR data for creating Random Forest and GBM-based risk stratification models. The model achieved an AUROC of 0.84 using structured data; however, unstructured clinical notes were not used in the model, limiting the predictive level [18]. In general, these works demonstrate the strong potential of ML techniques, particularly KNN, toward EC prediction. However, problems such as region-bounded datasets, exclusion of unstructured data, and lack of integration between data types remain common pitfalls.

III. Material and Methods

The four primary steps in the model suggested for the prediction of esophageal cancer are data preprocessing, feature selection, splitting the data, and classification. The process began with removing the irrelevant identifier columns since it was not of any predictive value. Missing and inconsistent values were treated next. Columns with more than 50% missing values were eliminated, and other missing values were filled with appropriate statistical methods, mean for continuous variables, mode for discrete variables, and 'Not Available' for categorical variables. Categorical values were encoded to ensure compatibility with machine learning algorithms. Feature selection was obtained by utilizing the Gini Index with a threshold value of 0.01 to select and retain the most relevant features for EC prognosis. The preprocessed dataset was then split into training (80%) and testing (20%) subsets. Seven different machine learning and deep learning models were trained and tested: Naive Bayes (NB), KNN, Logistic Regression (LR), AdaBoost, Linear Regression (LinR), Decision Tree (DT), and Multi-layer Perceptron (MLP). The experimental sample consisted of 3,985 patient records and 85 clinical and demographic features. Fig. 1 depicts the end-to-end pipeline data preprocessing and feature extraction, model training, testing, and final selection.

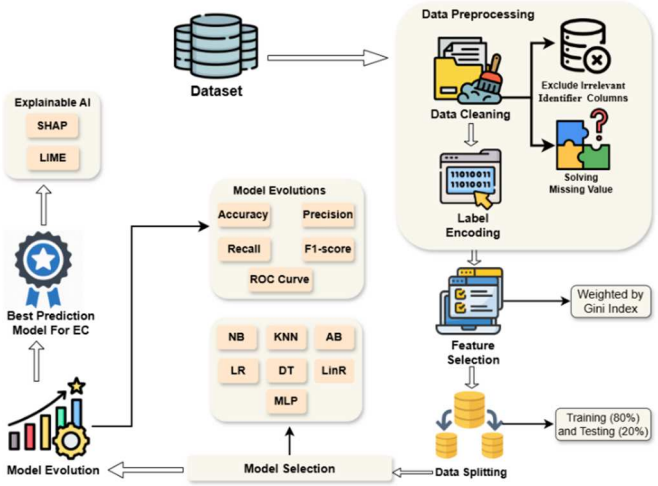


Fig. 1. System Architecture of the Proposed Machine Learning Based Esophageal Cancer Prediction Model

A. Dataset

The dataset was gathered from Kaggle [19] and includes 3,985 entries with 85 features collected from 2012 to 2015. Among them, 15% are female, and 85% are male. According to region, the United States accounts for 45% of the data, Vietnam for 21%, and other regions for 34% (1,358 items). The annual distribution has 120 entries from 2012, 1,671 from 2013, 2,094 from 2014, and 100 from 2015. The records have demographic, clinical, and lifestyle characteristics that are utilized to forecast the binary esophageal cancer outcome.

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B. Data Pre-processing

Preprocessing the data is the first and most crucial stage in creating a machine learning model because these models can process raw data. Clean, prepared data is not always available in a dataset. It is crucial to make sure that the data is appropriately cleaned, normalized, and converted to eliminate noise and handle missing values before performing any operations on the machine learning model in order to enhance the model's performance and accuracy. This study reduced the number of features from 85 to 81 (0 to 80) by removing the irrelevant identifiers from the dataset. When columns with more than 50% missing values were eliminated, 58 columns were left. The completed dataset consists of 42 object-type columns, 10 float64, and 6 int64 columns. After that, two main data categories, numerical and objective, were discovered. Object data also has two types: categorical and non-categorical. And numerical data also two types: continuous and discrete. "Not Available" was utilized to complete the missing values for both categorical and non-categorical objective data. While the mode value of the corresponding column was used to fill in discrete variables, the mean value of the corresponding column was used to fill in continuous variables for both continuous and discrete numerical data. Additionally, rows where the value in the column person_neoplasm_cancer_status was "Not Available" were removed from the dataset. Level encoding was applied to the dataset's object type data to transform numerical values into categorical values after filling in the missing values. This step guarantees compatibility with machine learning techniques, making the dataset more usable for future modelling tasks.

C. Feature Selection

Feature selection is one of the crucial steps in model development, which helps in enhancing model efficiency, reducing overfitting, and improving the generalization capability of the model [20]. In this study, the Gini Index (GI) was used to understand the contribution of each feature in differentiating between classes. The GI estimates the class distribution mixture or purity concerning a given feature and is based on the class probabilities and is computed using (1) [21].

$$\mathrm{GINI} = 1 - \sum_{\mathrm{i}=1}^{\mathrm{a}} P_{\mathrm{i}}^2 \qquad (1)$$

The total number of classes is a. P_i is the percentage or likelihood of the $i-th$ class. Fig. 2 and Table I. shows the probability distribution from GI used to measure the significance of the top feature linked to EC. A cut-off value of 0.01 was set for the GI to highlight relevant features. This technique increased efficiency and accuracy by identifying and eliminating less important features from the dataset.

TABLE I. MOST CONTRIBUTING FEATURES FOR MODEL PREDICTION BASED ON GINI_IMPORTANCE VALUES

Most Importance Features Identified by Gini Index		
Index	***Feature***	***Gini_Importance***
1	days_to_last_followup	0.301064
2	has_new_tumor_events_information	0.255939
3	primary_pathology_lymph_node_examined_count	0.115037
4	stage_event_tnm_categories	0.093969
5	tissue_source_site	0.039034
6	year_of_form_completion	0.032512
7	primary_pathology_age_at_initial_pathologic_diagnosis	0.031279
8	stage_event_pathologic_stage	0.025751
9	primary_pathology_initial_pathologic_diagnosis_method	0.017325
10	primary_pathology_number_of_lymphnodes_positive_by_he	0.016469
11	vital_status	0.014412
12	primary_pathology_neoplasm_histologic_grade	0.013678
13	days_to_birth	0.012063
14	frequency_of_alcohol_consumption	0.011719
15	has_radiations_information	0.010530

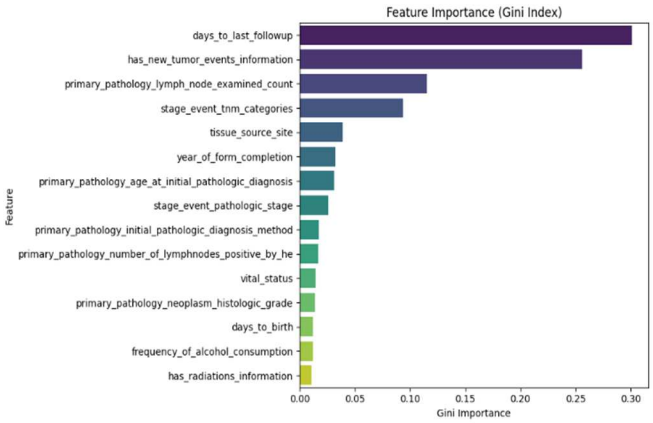


Fig. 2. Visualization of Feature Importance Based on the Gini Index

D. Data Splitting

To evaluate the performance of the proposed prediction models, the dataset was split into two subsets: training and testing. Especially, 80% of the data was used to train the machine learning models, and the remaining 20% was reserved for testing their predictive validity.

E. Model Selection

Seven different machine learning and deep learning models were used in this study to predict esophageal cancer: Naive Bayes, KNN, Logistic Regression, AdaBoost, Linear Regression, Decision Tree, and MLP. These classifiers were selected based on their effectiveness in classifying problems, especially medical diagnosis. All the models were trained using default hyperparameters without requiring any tuning to make things simple and have an honest baseline comparison. The goal was to compare the predictive performance of each of the models and find out the most suitable method for early EC detection.

K-Nearest Neighbors is a non-parametric algorithm that is simple and classifies a test sample by comparing it with the K nearest training instances. KNN does not build a model during training; instead, classification is performed for each query. The similarity between samples is measured using the Euclidean distance in (2).

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$$d(x_i, x_j) = \sqrt{\sum_{m=1}^{M} (x_{im} - x_{jm})^2} \quad (2)$$

In where x_i and x_j are sample vectors, M is the number of features, and x_{im} and x_{jm} are values in the $m-th$ feature. The less the distance, the more similar, and the last category is the one that gets the majority in voting between the nearest neighbors [22].

F. Model evolutions

Several performance measures, such as accuracy, recall, precision, f1 score, and ROC curve derived from the confusion matrix, were utilized to evaluate and compare the performance of ML algorithms. To compare and assess the power of machine learning algorithms, the area under the receiver operator characteristics (ROC) was assessed in this study.

Explainable AI methods were used to enhance the interpretability of the model. SHAP was applied for global and local feature contributions, while LIME provided instance level explanations for individual predictions. These combined approaches have increased transparency and strengthened confidence in the decision-making process of the model [24].

IV. Result Analysis

Performance of the machine learning classifiers was evaluated using accuracy, precision, recall, and F1-score, as shown in Table II. In all the traditional ML models, the best performance was obtained by KNN, which attained an accuracy of 96.84% and a very consistent range of high precision, recall, and F1-scores on both the non-cancer and cancer classes. Decision Tree and AdaBoost models obtained competitive results with an accuracy of 94.65% and 94.52%, respectively, with quite well-balanced metrics over the classes. Naive Bayes gave a moderate performance with an accuracy of 85.06%, whereas Linear Regression managed to perform at 88.08%, reflecting reasonable discriminative ability across classes. Logistic Regression performed comparatively lower, recording 71.78% accuracy with weaker recall and F1-score for the cancer class. In deep learning, the MLP classifier achieved an accuracy of 74.38%, with a considerably lower recall for the non-cancer class. The overall results show that the KNN model is the best when it comes to this dataset, but the Decision Tree and AdaBoost also give very strong and reliable performances concerning esophageal cancer prediction.

A. Confusion matrix

Fig. 3 demonstrates a performance analysis of the ML model through a confusion matrix evaluation. The performance metrics of accuracy, precision, recall, F1-score and specificity can be determined using the confusion matrix, which is displayed in Table II.

Here, a 2×2 confusion matrix is used, and performance is evaluated using accuracy, recall, precision, and F1-score defined in (3) – (6) [23].

$$Accuracy = \frac{AP + AN}{AP + PN + PP + AN} \quad (3)$$

$$Recall = sensitivity = \frac{AP}{AP + PN} \quad (4)$$

$$Precision = \frac{AP}{AP + PP} \quad (5)$$

$$F1score = 2 * \frac{precision * recall}{precision + recall} \quad (6)$$

AP: Actual positive, AN: Actual negative, PN: Predict negative, PP: Predict positive

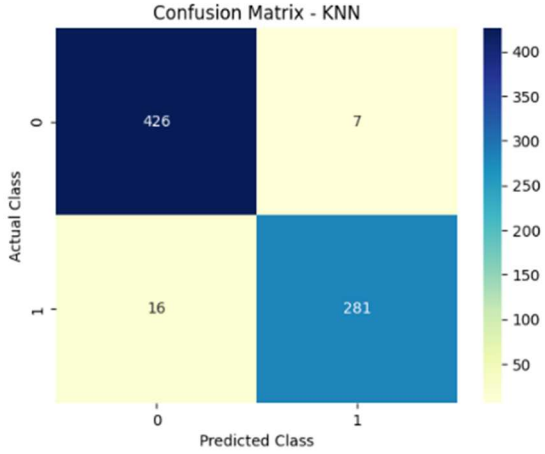


Fig. 3. Confusion Matrix of the KNN Model

TABLE II. Summary of Performance Metrics for Different Models for Esophageal Cancer Detection

Performance of Different Models					
Model	***Accuracy (%)***	***Class***	***Precision***	***Recall***	***F1-Score***
Naive Bayes	85.06	0	0.88	0.87	0.87
		1	0.81	0.82	0.82
KNN	96.84	0	0.96	0.98	0.97
		1	0.98	0.95	0.96
Logistic Regression	71.78	0	0.71	0.89	0.79
		1	0.75	0.46	0.57
AdaBoost	94.52	0	0.93	0.98	0.95
		1	0.97	0.90	0.93
Linear Regression	88.08	0	0.86	0.95	0.90
		1	0.92	0.77	0.84
Decision Tree	94.65	0	0.95	0.96	0.96
		1	0.94	0.92	0.93
MLP	74.38	0	0.89	0.64	0.75
		1	0.63	0.89	0.74

B. Roc-Curve

In the Fig. 4 presents the ROC curves for all models. KNN presents the largest AUC 1.00, closely followed by AdaBoost with 0.99 and Decision Tree with 0.98. Conversely, Naive Bayes, Linear Regression, and MLP present a moderate performance. Logistic Regression has the weakest record of AUC. Overall, the ROC results bring out KNN as the most successful model.

C. Explainable AI (XAI) Analysis

SHAP and LIME were applied to the top-performing KNN model to improve interpretability, offering both global feature importance and detailed instance-level explanations. These analyses provide clearer insights into how the model makes

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decisions and strengthen its transparency for clinical interpretation.

Fig. 5 the SHAP summary plot shows the global view of the impact that the selected top features have on the predictions made by the KNN model. Among the strongest contributors are features such as days_to_last_followup, days_to_birth, and stage_event_tnm_categories. Each of them pushes the predictions in different directions for higher and lower values. The distribution of SHAP values across the instances indicates how each feature contributes to the output of the model, its relative importance, and the direction of its influence.

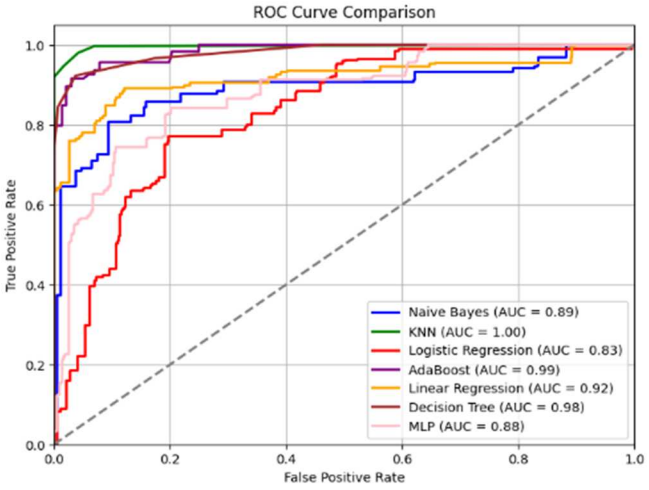


Fig. 4. Multiple Machine Learning Classifiers Analysis By ROC Curve

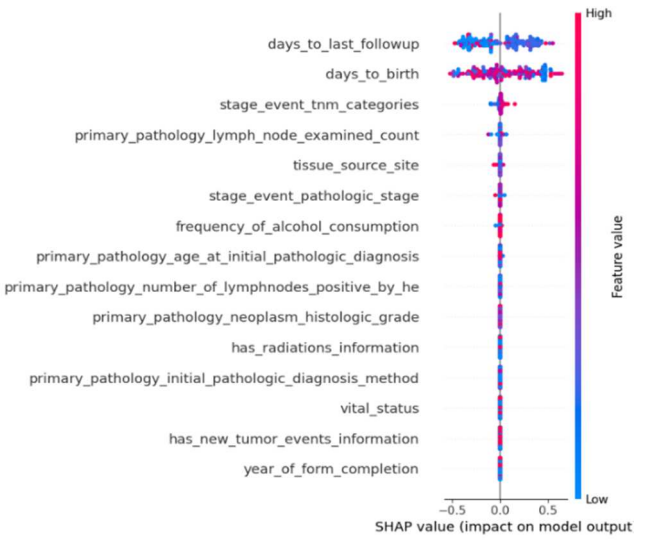


Fig. 5. SHAP Summary Plot Showing Global Feature Impact for the KNN Model

Fig. 6 presents SHAP waterfall plot of the KNN model prediction for a chosen test instance 0. The plot demonstrates how each feature contributes to the final prediction of the model. Features such as days_to_birth and days_to_last_followup represent the highest positive contribution in pushing the prediction toward the class of cancer, while other features have a very negligible or negative contribution. This plot provides a clear, instance-level view of the factors driving the model's decision.

Fig. 7 shows LIME explanation for test instance 0. It shows the features that most contributing according to the KNN model's prediction. The model predicts this instance as a cancer class with high confidence, and LIME lists days_to_birth and days_to_last_followup as strong contributors that drive the prediction towards class 1. The rest of the features, like tissue_source_site and vital_status, have a smaller or opposite effect. This instance-level interpretation provides clear insight into the factors shaping the model's decision.

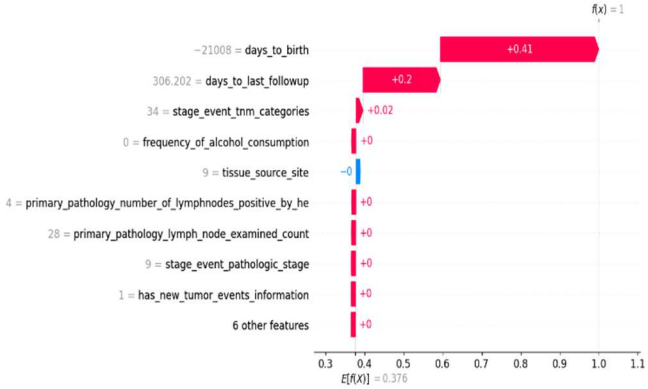


Fig. 6. SHAP Waterfall Explanation of KNN Model for Test Instance 0

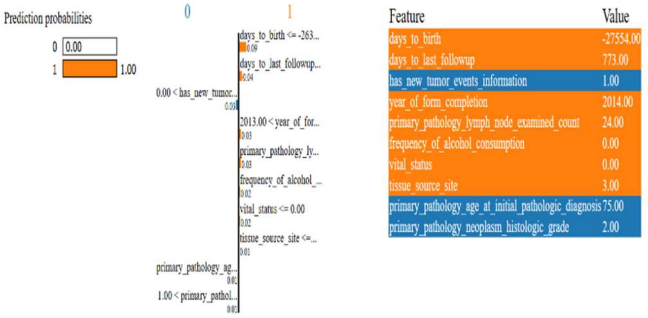


Fig. 7. LIME Explanation of Feature Contributions for KNN Prediction (Instance 0)

D. Discussion

This paper describes the machine learning-based framework for early esophageal cancer diagnosis on a large clinical dataset comprising 3,985 patient records. Amongst all models tested in this study, KNN produced the best accuracy, 96.84%, closely followed by the Decision Tree and AdaBoost models. SHAP and LIME provided complementary global and local explanations, consistently picking features such as days_to_last_followup, stage_event_tnm_categories, stage_event_tnm_categories, and days_to_birth as key predictors.

Compared with existing studies in Table III, such as Chen et al. [8] (81%), Etemadi et al. [10] (77%), Yoon et al. [15] (80.8%), and Zhang et al. [18] (84%), the proposed KNN model yields higher predictive accuracy. While much prior research used logistic regression, multivariate modeling, or imaging-based deep learning, our results indicate that well-tuned ML methods applied to structured clinical data yield higher

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diagnostic performance. However, the study has regional and temporal limitations of the dataset and lacks imaging or genomic features. These limitations drive the future research directions that are identified in the Future Work section. Overall, our results confirm that accurate and interpretable ML models can be helpful for early esophageal cancer detection, thus supporting data-driven clinical decision-making.

TABLE III. COMPARING THE PERFORMANCE OF OUR PROPOSED MODEL WITH EARLIER RESEARCH

Comparison of our proposed model with existing studies			
Author Name	***Dataset***	***Method***	***Accuracy or AUROC (%)***
Chen et al. [8]	Retrospective data with various risk factors	Logistic regression	81
Wang et al. [9]	Epidemiological data	Competing risk regression	76
Etemadi et al. [10]	Regional data with specific risk factors	Multivariate logistic regression	77
Raoof Nopour [13]	3,256 cases from Iranian hospitals	RF, XGBoost, KNN, SVM, ANNs	AUROC 92 (internal) and 83(external)
Rahman et al. [14]	Clinical and histological data	RF, XGBoost, ELR ensemble	80.50
Yoon et al [15]	232 NACRT cases	LR, SVM ensemble	80.8
Gong et al. [16]	10,588 EC cases from SEER database	XGBoost, ANN, LightGBM, RF, CatBoost	AUROC 85.2
Lin et al. [17]	Narrowband and white-light endoscopic images	U-Net + ResNet	85
Zhang et al. [18]	EHR data	Random Forest, GBM	84
Our Proposed Model	***Esophageal Cancer Dataset***	***KNN***	***96.84***

V. CONCLUSION

This work presents a strong machine learning model for early esophageal cancer diagnosis from high-volume clinical and demographic features. Following stringent data preprocessing and feature selection using the Gini Index, a comparison of all classifiers was performed and KNN was identified to have the highest accuracy of 96.84% and ideal precision, recall, and F1-score. The very high precision of the model marks its potential for precise risk stratification and early diagnosis in clinical practice. In addition, SHAP and LIME were used to offer both global and instance level explanations, helping make the model's predictions more transparent and easier to trust in a clinical setting. Future directions are external validation on larger multi-centers and temporally distinct test sets to validate robustness against distribution shifts. The methodological improvements will include multimodal imaging and genomic characteristics, calibrated uncertainty, and ensemble learning to enhance reliability and generalizability. Clinical translation will be involved in implementing a decision-support solution to support prioritization of screening and early risk triage with prospective assessment and workflow coordination.

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